

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
 Department of Computer Science and Engineering  
**ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING**

|                  |                                                                                                                |               |                                                |
|------------------|----------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------|
| Course Code:     | CSA06                                                                                                          | Course Title: | Design and Analysis of Algorithms              |
| CO Assessed:     | CO2 – Manipulation (BL1, BL2, BL3)                                                                             | Assessment:   | Assessment Tool 1 – Analytical Problem Solving |
| Weightage:       | 40% of CIA                                                                                                     | Total Marks:  | 100                                            |
| CO2 Statement:   | Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3) |               |                                                |
| Student Name:    | Manoj Kumar K                                                                                                  | Reg. No.:     | 192521041                                      |
| Section / Batch: | Information Technology(IT)                                                                                     | Date:         | 24/07/2026                                     |

**Instructions to Students**

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

| Question Type               |                                                                                                    | Questions   | Marks            |
|-----------------------------|----------------------------------------------------------------------------------------------------|-------------|------------------|
| Application-Based Questions | Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis | Q1<br>Q2,Q3 | 40<br>60 (2X30)  |
| <b>GRAND TOTAL:</b>         |                                                                                                    |             | <b>100 Marks</b> |

**SECTION A – APPLICATION BASED QUESTIONS**

**Q1. Linear Search – Element Not Present**

Given the unsorted dataset [17, 29, 43, 56, 68, 71], apply Linear Search to find the element 50. Clearly interpret the problem and explain the search process step-by-step. Count the total number of comparisons made before concluding that the element is absent. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the search is unsuccessful.

**Q2. Convex Hull – Brute Force Approach**

Apply the Brute Force Convex Hull algorithm on the set {(0,3), (2,2), (1,1), (2,1), (3,0), (0,0), (3,3)}. Interpret the problem and explain the concept of convex hull. Check all possible line pairs and identify valid hull edges. Show the logical steps used to determine the boundary. Count the total comparisons performed. Analyze the time complexity of the method. Interpret the result and its practical significance.

**Code of Conduct**

*I Manoj Kumar K(192521041) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*



Signature of the Student: \_\_\_\_\_

## RUBRICS FOR CALCULATION

| Criteria                         | Wt. | Level 4 – Exemplary                                                        | Level 3 – Proficient                                  | Level 2 – Developing                           | Level 1 – Beginning                                    |
|----------------------------------|-----|----------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Understanding of Problem Context | 20% | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps                    | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts          | 30% | Accurately applies appropriate concepts to solve complex problems          | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy   | Fails to apply appropriate concepts                    |
| Problem-Solving Approach         | 20% | Logical, well-structured, and efficient approach                           | Mostly logical approach with minor gaps               | Basic approach with some inconsistencies       | Illogical or unclear approach                          |
| Accuracy of Solution             | 20% | Completely correct solution with proper justification                      | Mostly correct with minor errors                      | Partially correct solution                     | Incorrect or incomplete solution                       |
| Clarity                          | 10% | Clear, well-organized, and properly explained solution                     | Understandable with minor clarity issues              | Limited clarity and explanation                | Poorly presented and difficult to understand           |

## Marks Evaluation:

| Question No.        | Understanding of Problem Context (20%) | Application of Concepts (30%) | Problem-Solving Approach (20%) | Accuracy of Solution (20%) | Clarity (10%) | Total Marks |
|---------------------|----------------------------------------|-------------------------------|--------------------------------|----------------------------|---------------|-------------|
| Q1<br>(40 Marks)    | 4                                      | 4                             | 4                              | 4                          | 4             | 40          |
| Q2<br>(30 Marks)    | 4                                      | 4                             | 4                              | 4                          | 4             | 30          |
| Q3<br>(30 Marks)    | 4                                      | 3                             | 3                              | 2                          | 2             | 22          |
| Overall Total (100) |                                        |                               |                                |                            |               | 92          |

Faculty In-charge

Course Coordinator

Head of Department

Signature: \_\_\_\_\_

Date: \_\_\_\_\_

Signature: \_\_\_\_\_

Date: \_\_\_\_\_

Signature: \_\_\_\_\_

Date: \_\_\_\_\_



Name: MANOJ KUMAR K

RegNo: 192521041

## CO2 - ASSESSMENT TOOL-1

Course: Design and Analysis of Algorithm

Course code: CSA0614

### Linear Search - Element Not Problem

Given

$\Rightarrow$  Dataset = [17, 29, 43, 56, 68, 71]

Search Key = 50

Problem Interpretation

The dataset is unsorted so Linear Search is the appropriate algorithm. The algorithm compares the key with each element sequentially until the element is found or all elements have been checked.

Search Analysis

\*  $50 \neq 17$

\*  $50 \neq 29$

\*  $50 \neq 43$

\*  $50 \neq 56$

\*  $50 \neq 68$

\*  $50 \neq 71$

Since no element matches so, the search is unsuccessful.

Comparison Count

\* Total Count = 6



## Complexity Analysis

\* Best Case:  $O(1)$

\* Average Case:  $O(n)$

\* Worst Case:  $O(n)$

## Correctness Verification

$\Rightarrow$  Every element was checked exactly once and one match was found. Therefore, the algorithm correctly concludes that 50 is not present in the dataset.

## Efficiency Interpretation

An unsuccessful linear search requires checking all elements, hence the running time is  $O(n)$ , making it suitable for small or unsorted datasets but inefficient for large datasets.

## Final Answer

Element Found: No

Total Comparisons: 6

Time Complexity:  $O(n)$



## Convex Hull (Brute Force)

### Problem Interpretation

Find the Smallest Convex Polygon enclosing the points using the Brute Force Convex Hull algorithm

### Analysis

Check every pair of points. If all other points lie on the same side of the line, the pair is a hull edge

### Convex Hull Vertices

$(0,0), (3,0), (3,3), (0,3)$

### Hull Edges

\*  $(0,0) \rightarrow (3,0)$

\*  $(3,0) \rightarrow (3,3)$

\*  $(3,3) \rightarrow (0,3)$

\*  $(0,3) \rightarrow (0,0)$

### Comparisons

\* Point Pairs = 21

\* Total Comparison =  $21 \times 5 = 105$

### Time Complexity

$$O(n^3)$$

### Result Interpretation

The four corner points form the Convex hull. The remaining points lie inside the boundary

## Conclusion

The Brute Force algorithm correctly finds the Convex hull with less comparison and  $O(n^3)$  time complexity. It is simple but inefficient for large dataset.